Global Academy of Technology, Bengaluru

Department of Electronics and Communication Engineering

Question Bank

Engineering Electromagnetics - BEC24403

Module 1

Question No.	Questions	Marks
1	State and explain Coulomb's Law of force between two point charges in vector form and also indicate the units of quantities in the force equation.	**6**
2	Define electric field intensity. Derive an expression for field at point due to a point charge.	**6**
3	Explain different types of charge distributions with mathematical expressions.	**5**
4	Derive the expression for Electric Field intensity at a point due to N point charges.	**6**
5	Derive the expression for Electric field intensity at a point due to infinite line charge .	**6**
6	Derive the expression for Electric field intensity at a point due to infinite sheet charge	**6**
7	Define Electric Flux Density. Find the relation between Electric Flux Density and Field Intensity	**5**
8	Let a point charge Q_1 = 25 nC be located at $P_1(4, -2, 7)$ and a charge Q_2 = 60 nC be at $P_2(-3, 4, -2)$. find **E** at $P_3(1, 2, 3)$.	**6**
9	Two point charges of magnitude 2mC and -7 mC are located at places P1(4,7,-5), and P2(-3,2,-9) respectively in free space, find the vector force on charge at P2.	**6**
10	Calculate **D** at point $P(6,8, -10)$ produced by: (*a*) a point charge of 30mC at origin.(*b*) a uniform line charge 40 μC/m on the *z*-axis. c) Uniform surface charge density ρ_s=57.2μC/m^2 on the plane x=9. d) due to all these charge configurations.	**8**
11	Find the total charge inside the volume with density $\rho_v = 10z^2e^{-10x}\sin\pi y$ where $-1\leq x\leq 2$, $0\leq y\leq 1$ and $3\leq z\leq 4$.	**6**
12	Two point charges of 5 μC and -3μC are placed along straight line 10m apart. Determine the location of a third charge of 4μC such that it is subjected to no force.	**6**
13	A uniform line charge 25nc/m lies on the line x=-3m and y=4m in free space. Find the electric field intensity at a point (2,3,15)m	**6**
14	Find the electric field at p(1,1,1) caused by four identical 3nC charges located at $p_1(1,1,0)$, $p_2(-1,1,0)$, $p_3(-1,-1,0)$ and $p_4(1,-1,0)$	**6**
15	Find the total charge in the volume where ρ_v is given by $\rho_v = \rho^2 z^2 \sin(0.6\phi)$ C/m^3 ,$0 \leq \rho \leq 0.1, 0 \leq \emptyset \leq \pi$, $2 \leq z \leq 4$	**6**
16	An infinite uniform line charge ρ_L= 2 nC/m lies along the *x* axis in free space, while point charges of 8nC each are located at (0,0,1) and (0,0,-1).a) Find **E** at (2,3,-4).	**6**
17	Calculate **D** in rectangular coordinates at point $P(2,-3, 6)$ produced by: (*a*) a point charge Q_A= 55 mC at $Q(-2, 3,-6)$; (*b*) a uniform line charge ρ_{LB}= 20 mC/m on the *x* axis.	**6**

18	Uniform sheet of charge of $25nC/m^2$ lies in the plane z = 0. A point charge of $6\mu C$ is located at origin and uniform line charge of 180nC/m lies along x axis. Find **D** at A(0,0,4).	**5**
19	$\rho_v = (x + 2y + 3z)nC/m^3, 1 \le x \le 2,\ -2 \le y \le 1, 2 \le z \le 3,$ find the total charge enclosed.	**6**
20	A uniform volume charge density of $0.2\mu C/m^3$, is present throughout a spherical shell, extending from r = 3cm to 5cm. if ρ_v=0, elsewhere find total charge present throughout the spherical shell.	**6**
21	Three infinite uniform sheet of charge are located in the free space as follows: $3nC/m^2$ at z = -4, $6nC/m^2$ at z =1, $-8nC/m^2$ at z = 4. Find **E** at $P_A(2, 5,-5)$	**6**
22	Find **E** at (1,5,2) if point charge of $6\mu C$ is located at Q(0,0,1), uniform line charge of 180nC/m lies along x axis and uniform sheet of charge of $25nC/m^2$ lies in the plane z = -1.	**8**

Module 2

Question No.	Questions	Marks
1	State and prove Gauss Law	6
2	Given $\mathbf{D}=0.3r^2\mathbf{a}_r nC/m^2$ in free space: i)Find **E** at P(2, 25^0,90^0). ii)Find the total charge within the sphere r=3.	6
3	Apply Gauss law to derive for the field of point charge.	6
4	Using Gauss law find the field of line charge.	6
5	Apply Gauss law for differential volume element and hence derive Maxwell's first equation.	12
6	Derive Maxwell's first equation.	5
7	State and prove Divergence Theorem.	5
8	Given a 60-μC point charge located at the origin, find the total electric flux passing through that portion of the sphere r = 26 cm bounded by $0 < \theta < \pi/2$ and $0 < \varphi < \pi/2$	6
9	Given a 60-μC point charge located at the origin, find the total electric flux passing through the closed surface defined by ρ = 26 cm and $z = \pm 26$ cm.	6
10	In free space, let $\mathbf{D} = 8xyz^4\mathbf{a}_x + 4x^2z^4\mathbf{a}_y + 16x^2yz^3\mathbf{a}_z$ pC/m^2. (*a*) Find the total electric flux passing through the rectangular surface $z = 2$, $0 < x < 2$, $1 < y < 3$, in the $\mathbf{a}_z$ direction. (*b*) Find **E** at P (2, −1,3). (*c*) Find an approximate value for the total charge contained in an incremental sphere located at *P*(2,−1, 3) and having a volume of 10^{-12} m^3.	8
11	Find a numerical value for div **D** at the point specified: $\mathbf{D} = (2xyz - y2)\mathbf{a}_x + (x^2z - 2xy)\mathbf{a}_y + x^2y\mathbf{a}_z C/m^2$ at $P_A(2, 3, -1)$.	6
12	Find a numerical value for div **D** at the point specified: $\mathbf{D} = 2\rho z^2 \sin^2\varphi\ \mathbf{a}_\rho + \rho z^2 \sin(2\varphi)\mathbf{a}_\varphi + 2\rho^2 z \sin^2\varphi\ \mathbf{a}_z C/m^2$ at $P_B(\rho = 2, \varphi = 110^\circ, z = -1)$.	7
13	Determine an expression for the volume charge density associated with $\mathbf{D} = 4xy/z\mathbf{a}_x + 2x^2/z\mathbf{a}_y - 2x^2y/z^2\mathbf{a}_z$	6
14	Evaluate both sides of the divergence theorem for the field $\mathbf{D} = 2xy\mathbf{a}_x + x^2\mathbf{a}_y C/m^2$ and the rectangular parallelepiped formed by the planes x = 0 and 1, y = 0 and 2,and z = 0 and 3.	10
15	Given the field $\mathbf{D} = 6\rho \sin(\varphi/2)\mathbf{a}_\rho + 1.5\rho \cos(\varphi/2)\mathbf{a}_\varphi C/m^2$, evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\varphi = 0$, $\varphi = \pi$, $z = 0$, and $z = 5$.	10
16	Let $\mathbf{D} = 4xy\mathbf{a}_x + 2(x^2 + z^2)\mathbf{a}_y + 4yz\mathbf{a}_z nC/m^2$ and evaluate surface integrals	10

	to find the total charge enclosed in the rectangular parallelepiped $0 < x < 2$, $0 < y < 3$, $0 < z < 5$ m.	
17	Prove that work done is independent of the path carried between initial and final points.	**6**
18	Find the work done in moving a charge of 5μC from origin to the point P (2, -1,4) through the field E= $2xyz\mathbf{a_x}+x^2z\mathbf{a_y}+x^2y\mathbf{a_z}$ v/m via the paths i) straight line segments from (0,0,0) to (2, -1,4) ii) straight line x= -2y and z=2x iii) curve $x=-2y^3$ and $z=4y^2$.	**10**
19	Develop a formula for the potential field of system of charges.	**6**
20	An electric field is expressed in terms of rectangular coordinates by $\boldsymbol{E}=6x^2\boldsymbol{a_x}+6y\boldsymbol{a_y}+4\boldsymbol{a_z}\ v/m$ Find: (*a*) V_{MN} if points M and N are specified by $M(2, 6,-1)$and $N(-3,-3, 2)$; (*b*) V_M if $V = 0$ at $Q(4,-2,-35)$; (*c*) V_N if $V = 2$ at$P(1, 2,-4)$.	**6**
21	Each coordinate axis carries uniform line charge density of 20nC/m, in free space. Given the points A(1,2,3) and B(6,8,10). Find V_{AB} due to i)ρ_L on the x axis ii) ρ_L on the y axis iii) ρ_L on the z axis iv) all of the above.	**7**
22	In the free space, a line charge of ρ_L=80nC/m lies along entire z axis, while point charges of Q=100nC each at (1,0,0) and (0,1,0). Find V_{PQ}, given P(2,1,0) and Q(3,2,5).	**7**
23	An electric field is expressed as $\boldsymbol{E}=6x^2\boldsymbol{a_x}+6y\boldsymbol{a_y}+4\boldsymbol{a_z}\ v/m$ Find i)V_{MN} if the points M and N are specified by M(2,6,-1) and N(-3,-3,2). ii)V_M if V=0 at Q(4,-2,-35). iii)V_N if V=2 at P(1,2,-4).	**6**
24	A 15nC point charge is at the origin in the free space. Calculate V_1, if the point P_1 is located at P_1(-2,3, -1) and i) V=0 at ∞ ii) V=0 at (6,5,4) iii) V=5v at(2,0,4).	**6**

Module 3

Question No.(Unit)	Questions	Marks
1(2)	State Biot-Savart law and provide its mathematical expression.	**6**
2(2)	Deduce an expression for the field of infinite line carrying direct current of I amper by using B-S law.	**6**
3(2)	State Amper's circuital law and hence calculate field of infinite line carrying direct current of I ampere.	**6**
4(2)	Determine the field of coaxial conductor.	**8**
5(2)	Prove that $\boldsymbol{\nabla X H = J}$.	**8**
6(2)	Evaluate $\boldsymbol{\nabla X G}$ at P_A(4,30°,45°) if G$G = \boldsymbol{\sin\theta}(\boldsymbol{a_r}+\boldsymbol{a_\theta}+\boldsymbol{a_\phi})$	**6**
7(2)	Prove Stoke's theorem.	**6**
8(2)	A solid conductor of circular cross-section with a radius of 5 mm has a conductivity that varies with radius. The conductor is 20 m long and there is a potential difference of 0.1 V dc between its two ends. Within the conductor,$\boldsymbol{H}=10^5\rho^2\boldsymbol{a}_\phi\ A/m$ a) find σ as a function of ρ. b) What is the resistance between the two ends?	**6**
9(2)	Given the field $\mathbf{H} = 20\rho^2\mathbf{a}_\varphi$A/m. a) Determine the current density **J** b) Determine the total current through surface integration, $\rho = 1$, $0 < \varphi < 2\pi$, $z = 0$, c) Find the total current once more through line integral.	**6**

	An infinitesimal length 10^{-3} m of wire is located at the point(1,0,0) and carries a current 2A in the direction of the unit vector ax. Find the magnetic field intensity due to current element at the point (0,2,2)	
10(3)	A solid conductor has $H_{\emptyset} = 468.34\rho + 585.4 * 10^5\rho^3\ A/m$. Find the magnetic flux.,0 <= ρ <= 0.01, 0 < z < 1.	**6**
11(3)	$\boldsymbol{H} = \frac{20(x\boldsymbol{a_x} + y\boldsymbol{a_y})}{(x^2+y^2)} A/m$, show that $\boldsymbol{\nabla}.\boldsymbol{B} = 0$, Find **J**.	**6**
12(3)	Define scalar magnetic potential and prove that magnetic field is non conservative.	**6**
13(3)	Assume that $\mathbf{A} = 50\rho^2\mathbf{a}_z$Wb/m in a certain region of free space. a) Find **H** and **B**: b) Find **J** c)Use **J** to find the total current crossing the surface $0 \le \rho \le 1$, $0 \le \varphi < 2\pi, z = 0$.	**8**
	UNIT 1	
14(1)	$\boldsymbol{H} = (2\rho - \rho^2)\boldsymbol{a_\emptyset} A/m, \quad for \quad 0 \le \rho \le 1,$ Determine the current density. What is the total current passing through the surface $0 \le \rho \le 1, z = 0$ in $\mathbf{a_z}$ direction. Check by another method.	**6**
15(1)	Derive Continuity equation.	**6**
16(1)	$\boldsymbol{J} = \frac{4}{r^2}\cos\theta\boldsymbol{a_r} + 20e^{-2r}\sin\theta\boldsymbol{a_\theta} - r\sin\theta\boldsymbol{a_\emptyset}\ A/m^2$ i) find J at r=3,θ=0 and Φ =π ii) Find the total current passing through area r=3,0≤θ≤ 20^0 and 0≤Φ =2π in $\mathbf{a_r}$ direction.	**6**
17(1)	$\mathbf{H} = \frac{2}{\pi\rho}\left(1 + \frac{10^7\ \rho^3}{6}\right)\boldsymbol{a_\emptyset} + 8\,\boldsymbol{a_z}\ A/m, \quad for\ 0 \le \rho \le 0.01m$ $\mathbf{H} = \frac{16}{3}\ \pi\rho\boldsymbol{a_\emptyset} + 8\boldsymbol{a_z}\ for\ \rho \ge 0.01m$ a) Find **J** everywhere. b) What is I at ρ =0?	**6**
18(1)	The magnetic field intensity is given in a certain region of space as $\boldsymbol{H} = \frac{x+2y}{z^2}\boldsymbol{a_y} + \frac{2}{z}\boldsymbol{a_z}$ A/m. a) find **J**. b) Use **J** to find the total current passing through the surface z = 4, 1 < *x* <2, 3 < *y* <5, in the $\mathbf{a}_z$ direction	**6**
19(1)	Given $\boldsymbol{H} = -\boldsymbol{y}(\boldsymbol{x^2} + \boldsymbol{y^2})\boldsymbol{a_x} + \boldsymbol{x}(\boldsymbol{x^2} + \boldsymbol{y^2})\boldsymbol{a_y}$ A/m in the z=0 plane for $-5 \le x, y \le 5$. Find the total current passing through z = 0 plane in the $\mathbf{a_z}$ direction, inside the rectangle $-1 \le x \le 1, -2 \le y \le 2$	**6**

Module 4

Question No.(Unit)	Questions	Marks
1(1)	Derive an expression for magnetic force on a differential current element.	**8**
2(1)	Derive an expression for magnetic force between two differential current elements.	**8**
3(1)	State Lorentz's force equation and provide mathematical expression.	**8**
4(1)	The point charge *Q* = 18nC has a velocity of 5×10^6 m/s in the direction $\mathbf{a}_v = 0.60\mathbf{a}_x + 0.75\mathbf{a}_y + 0.30\mathbf{a}_z$. Calculate the magnitude of the force exerted on the charge by the field: (*a*) $\mathbf{B} = -3\mathbf{a}_x + 4\mathbf{a}_y + 6\mathbf{a}_z$ mT; (*b*) $\mathbf{E} = -3\mathbf{a}_x + 4\mathbf{a}_y + 6\mathbf{a}_z$ kV/m; (*c*) B and E acting together.	**10**

5(1)	The field $\mathbf{B} = -2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z$ mT is present in free space. Find the vector force exerted on a straight wire carrying 12 A in the $\mathbf{a}_{AB}$ direction, given $A(1, 1, 1)$ and (*a*) $B(2, 1, 1)$(*b*) $B(3, 5, 6)$.	**10**
6(1)	Two differential current elements, $I_1\Delta\mathbf{L}_1 = 3 \times 10^{-6}\mathbf{a}_y$ A・m atP_1 (1, 0, 0) and $I_2\Delta\mathbf{L}_2 = 3\times10^{-6}(-0.5\mathbf{a}_x + 0.4\mathbf{a}_y + 0.3\mathbf{a}_z)$ A・mat P_2 (2, 2, 2), are located in free space. Find the vector force exerted on: (*a*) $I_2\Delta\mathbf{L}_2$ by $I_1\Delta\mathbf{L}_1$ (*b*) $I_1\Delta\mathbf{L}_1$ by $I_2\Delta\mathbf{L}_2$.	**10**
7(1)	An electron is moving through constant velocity $\boldsymbol{V} = 4 * 10^7 \boldsymbol{a_x}\, m/s$ along negative x axis. At the origin it encounters a uniform magnetic field$\boldsymbol{B} = -4 * 10^{-3}\boldsymbol{a_y}\, T$, find the acceleration.	**8**
8(1)	Find the force in the following conditions: i) Conductor length of 4m along $\mathbf{a_y}$ direction, carrying a current of 10 ampere in the field of $\mathbf{B} = 0.05\mathbf{a_x}$ Tesla ii) Conductor length of 0.3m along a_z direction, carrying a current of 5 ampere in the field of $\mathbf{B} = 3.5\text{x}10^{-3}(\mathbf{a_x}\text{-}\mathbf{a_y})$ Tesla	**8**
	UNIT 2	
9(2)	Explain Faraday's law and hence obtain an expression for the same.	**8**
10(2)	Given $\boldsymbol{B} = 6\cos(10^6\, t)\sin(0.01x)\,\boldsymbol{a_z} mT$. Find the magnetic flux passing through the surface z = 0, 0 < x < 20m, 0 < y < 3m at t=1µs.	**8**
11(2)	A perfectly conducting filament containing a small 500-Ω resistor is formed into a square of 0.5m side. Find $I(t)$ if $\mathbf{B} = 0.3\cos(120\pi t - 30^\circ)\,\mathbf{a}_z$ T.	**8**
	iii)	
12(2)	List Maxwell's equations for time varying field in differential form and convert them to integral form.	**12**
13(2)	List Maxwell's equations for steady field in point form and convert them to integral form	**10**
14(2)	Modify Maxwell's 2nd and 3rd equations with respect to time varying fields.	**10**
15(2)	Explain conduction current and displacement current and prove that conduction current is equal to displacement current.	**12**
16(2)	For a certain material, σ=0, μ_r= 1, $\boldsymbol{E} = 800\sin(10^6 t - 0.01z)\,\boldsymbol{a}_y v/m.$ Use Maxwell's equations to find ε_r.	**12**
17(2)	Assume a homogeneous material of infinite extent with $\in = 2 * 10^{-10}\frac{F}{m}$, $\mu = 1.25 * 10^{-5}\frac{H}{m}$, $\sigma = 0, \boldsymbol{E} = 400\cos(10^9 t - kz)\,\boldsymbol{a}_y v/m$, if all the fields vary sinusoidally , use Maxwell's equations to find **D,B,H** and k	**12**
18(2)	A charge Q=5 * 10^{-18}C, is moving through a uniform magnetic field $B = -0.4\boldsymbol{a_x} + 0.2\boldsymbol{a_y} - 0.1\boldsymbol{a_z}\, T$, with a velocity $V = (2\boldsymbol{a_x} - 3\boldsymbol{a_y} + 6\boldsymbol{a_z})10^5 m/s$, i) What electric field is present at t=0, if the net force on electron is zero? ii) If the electric field intensity is entirely on $\mathbf{a_x}$ direction and $\lvert F_{net}\rvert$=2pN at t=0 find E_x.	**10**

Module 5

Question No.	Questions	Marks
1	What is Forward travelling wave and Backward travelling wave in free space?	8
2	State and prove Poynting theorem.	8
3	Explain depth of penetration along with relevant equations.	8
4	Derive for the E and H for uniform plane wave in free space propagation.	8

5	Derive for the E and H for uniform plane wave in good conductor.	**8**
6	Derive for the E and H for uniform plane wave in perfect dielectrics	**8**
7	Discuss wave propagation in lossy dielectrics	
7(PD)	A 300MHz wave propagating through fresh water assuming a lossless medium μ_r=1, $€_r$=78 (at 300MHz) find the β, v, **λ,** η. If E_0=0.1V/m, also find E_x, and H_y.	
8(PD)	Find propagation constant, attenuation constant, phase constant, intrinsic impedance in distilled water at ω = 10^{11} rad/s. in super high frequency (SHF) band the parameters are μr=1, €r=50 , σ = 20℧/m	
9(FS)	An E field in free space is given as E=800cos(10^8t- βy)a_z V/m. find (a) β (b) **λ (c) H at P(0.1, 1.5, 0.4) at t=8ns**	
10(FS)	A uniform plane wave in free space is given by E=(200 find (a) β (b) ω (c) f (d) **λ (e)** η at Z=8mm, t=6ps	
11(FS)	The electric field intensity of a 300MHz uniform plane in free space is given as Es=(20+j50) (ax+2ay) e-j βzV/m i) Find ω, λ, V and β ii) Find E at time t=1ns, z=10cm iii) What is $\|H\|_{max}$	
12(pD)	A wave propagating in a lossless dielectric has the components E=500cos(10^7t – βz)ax V/m and H=1.1cos(10^7t – βz)ay A/m. if the wave is travelling at V=0.5c find (i) μ_r (ii) €r (iii) β (iv) η (v) **λ**	
13(pd)	Let Es=(1000ax+400az)e-j10y v/m for a 250MHz uniform plane wave propagating in a perfect dielectric. If the maximum amplitude of H is 3A/m, fid (i) μr (ii) η (iii) **λ** (iv) v (v) €r	
14(pD)	A 1MHz uniform plane wave with an amplitude e of 25V/m is propagating in a material for which €r =4, μ_r=9 and σ=0. Propagation s in ax direction find (i)v (ii) **λ**(iii) β(iv) η	
15(pd)	A 9.375GHz uniform plane wave is propagating in polyethylene. If the amplitude of electric field intensity is 500V/m and the material is assumed to be lossless find (i) Phase constant (ii) Wavelength of polythene (iii) Velocity of propagation (iv) Intrinsic impedance (v) Amplitude of the magenetic field intensity	
9(conductor)	A steel pipe is constructed of a material for which μ_r = 180 and σ = 4 × 10^6 S/m. The two radii are 5 and 7 mm, and the length is 75 m. If the total current *I* (*t*) carried by the pipe is 8 cos *ωt* A, where *ω*= 1200 π rad/s, find: (*i*) the skin depth; (*ii*) the effective resistance; (*iii*) the dc resistance; (*iv*)the time-average power loss.	**8**
15(con)	Calculate the intrinsic impedance, propagation constant and wave velocity for conducting medium in which σ=58M S/m, μr=1, €r=1 at frequency of 10MHz.	
16(PD)	An amplitude of magnetic field intensity is 20A/m in ay direction wave is propagating az direction at a frequency of 2x10^9 rad/sec. find (i) **λ** (ii) f (iii) T and (iv) E_{max} (v) η	
17(LD)	Given a non-magnetic material having €r=2.25 and σ=10^{-4}℧/m find the numerical values at 2.5MHz for (i) loss tangent (ii) attenuation Constant (iii) the phase constant (iv) Intrinsic Impedance	